A scaling symmetry of the TorC background equations: why υ_2 is not a parameter, and why only ratios of the Ω's are measurable

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Abstract

Every term of the TorC background Friedmann equation is homogeneous of degree two in time derivatives. This implies an exact one-parameter scaling symmetry under which the three density parameters are multiplied by a common factor, H and $\dot{\varpi}$ by its square root, and $\varpi(a)$ is unchanged. The observable $E(z) = H/H(a{=}1)$ is invariant. Three apparently distinct facts are consequences of this single symmetry: (i) υ_2 does not affect background distances; (ii) $\Omega_{m,0}$, $\Omega_{r,0}$ and $\Omega_{\Lambda,0}$ are measurable only through their ratios; (iii) the posterior in the $(\Omega_{m,0}, \Omega_{\Lambda,0})$ plane is an elongated wedge pointing at the origin. All statements are verified numerically against the solver to $\sim 10^{-9}$.

1 Setup

The Lagrangian is

$$\mathcal{L} = \frac{3M_{\rm P}^2\upsilon_2}{4\varsigma}\left[\tfrac{1}{2}(4\varsigma-\varpi^2)R - g^{\mu\nu}\partial_\mu\varpi\,\partial_\nu\varpi + \varsigma\phi^2 - \varpi^2\phi^2 + \tfrac{4}{3}\sqrt{J_\mu J^\mu}\right] - M_{\rm P}^2\Lambda + \mathcal{L}_{\rm M}, \tag{1}$$

$$J_\mu \equiv \varpi^3\partial_\mu(\phi/\varpi) - \partial_\mu\phi. \tag{2}$$

Note for later that υ_2 occurs *once*, in the overall prefactor, and nowhere inside the bracket, in J_μ, or in $\mathcal{L}_{\rm M}$.

The background Friedmann equation is

$$H^2 = \frac{1}{\varpi^2-4\varsigma}\left[-\frac{4H_0^2\varsigma}{\upsilon_2}S(a) + 2H\varpi\dot{\varpi} + \dot{\varpi}^2 + \frac{4\big(H\varsigma(\varpi^2-1)+\varsigma\varpi\dot{\varpi}\big)^2}{\varsigma-\varpi^2}\right], \tag{3}$$

where we abbreviate the matter source

$$S(a) \equiv \Omega_{r,0}a^{-4} + \Omega_{m,0}a^{-3} + \Omega_{\Lambda,0}. \tag{4}$$

The three density parameters appear in (3) only inside $S(a)$, and $S(a)$ appears only once.

2 The observation: every term is quadratic in rates

Call H and $\dot{\varpi}$ *rates* (one time derivative each); H_0 is also a rate. Then in (3):

term	rates
H^2	2
$H_0^2 S(a)/\upsilon_2$	2
$2H\varpi\dot{\varpi}$	2
$\dot{\varpi}^2$	2
$4\big(H\varsigma(\varpi^2-1)+\varsigma\varpi\dot{\varpi}\big)^2/(\varsigma-\varpi^2)$	2

Every term carries exactly two. The quantities ϖ, ς and a appear at many powers but carry no time derivative and never spoil the count. Equation (3) is therefore homogeneous of degree 2 in rates.

3 The symmetry

Proposition. Let $\kappa > 0$ be an arbitrary constant — a free parameter labelling the transformation, with no physical meaning ($\kappa = 1$ is the identity). It is *not* the speed of light. Then the transformation

$$\begin{aligned} &\Omega_{r,0} \to \kappa^2\Omega_{r,0}, \qquad \Omega_{m,0} \to \kappa^2\Omega_{m,0}, \qquad \Omega_{\Lambda,0} \to \kappa^2\Omega_{\Lambda,0}, \\ &H(a) \to \kappa H(a), \qquad \dot{\varpi} \to \kappa\dot{\varpi}, \qquad \varpi(a) \to \varpi(a), \qquad \tau \to \tau/\kappa \end{aligned} \tag{5}$$

leaves (3) invariant.

Proof. Under (5), $S(a) \to \kappa^2 S(a)$ since every term of (4) is linear in the Ω's. Then term by term:

$$\begin{aligned} H^2 &\to \kappa^2 H^2, \\ \frac{4H_0^2\varsigma}{\upsilon_2} S &\to \kappa^2\,\frac{4H_0^2\varsigma}{\upsilon_2} S, \\ 2H\varpi\dot{\varpi} &\to (\kappa)(\kappa)\,2H\varpi\dot{\varpi} = \kappa^2\,2H\varpi\dot{\varpi}, \\ \dot{\varpi}^2 &\to \kappa^2\dot{\varpi}^2, \\ H\varsigma(\varpi^2-1)+\varsigma\varpi\dot{\varpi} \to \kappa\big[H\varsigma(\varpi^2-1)+\varsigma\varpi\dot{\varpi}\big] &\implies (\cdot)^2 \to \kappa^2(\cdot)^2, \end{aligned}$$

while the prefactor $1/(\varpi^2 - 4\varsigma)$ and the denominator $\varsigma - \varpi^2$ are unchanged because $\varpi(a)$ is unchanged. Both sides of (3) acquire exactly κ^2. □

3.1 Corollary: the observable is blind to it

The quantity compared with data is

$$E(z) \equiv \frac{H(z)}{H(a{=}1)} \;\longrightarrow\; \frac{\kappa\,H(z)}{\kappa\,H(a{=}1)} = E(z). \tag{6}$$

The common factor κ cancels between numerator and denominator. The symmetry multiplies H by the *same* factor at every epoch, and E is constructed precisely to divide out the overall rate.

3.2 Three equivalent statements of the same symmetry

Because the Ω's enter (3) only in the combination $H_0^2\Omega/\upsilon_2$, the following are the same transformation:

$$\Omega_i \to \kappa^2\Omega_i \quad\Longleftrightarrow\quad \upsilon_2 \to \upsilon_2/\kappa^2 \quad\Longleftrightarrow\quad \tau \to \tau/\kappa. \tag{7}$$

The last form is the statement that the symmetry is a reparametrisation of time: a rescaling of the clock, not of the physics.

4 Consequence 1: υ_2 is not a background parameter

By (7), changing υ_2 is a pure time rescaling. Writing $\tau = \sqrt{\upsilon_2}\,\sigma$ removes υ_2 from the equations of motion and from the early-time series initial conditions entirely, so $a(\sigma)$ and $\varpi(\sigma)$ are the same functions for every υ_2. Since $H \propto a^{-2}\,\mathrm{d}a/\mathrm{d}\tau$ and $\mathrm{d}a/\mathrm{d}\tau = a'(\sigma)/\sqrt{\upsilon_2}$, we obtain

$$H(a;\upsilon_2) = \frac{H(a;\upsilon_2{=}1)}{\sqrt{\upsilon_2}} \tag{8}$$

at *every* epoch, hence $E(z)$ is independent of υ_2.

This is visible in (1): υ_2 appears only in the overall prefactor $3M_{\mathrm{P}}^2\upsilon_2/4\varsigma$, so it is absorbed by the renaming $\widetilde{M_{\mathrm{P}}}^2 \equiv M_{\mathrm{P}}^2\upsilon_2$, $\widetilde{\Lambda} \equiv \Lambda/\upsilon_2$:

$$\mathcal{L} = \frac{3\widetilde{M_{\mathrm{P}}}^2}{4\varsigma}\big[\ \cdots\ \big] - \widetilde{M_{\mathrm{P}}}^2\widetilde{\Lambda} + \mathcal{L}_{\mathrm{M}}. \tag{9}$$

By contrast ς appears in the prefactor *and* in $(4\varsigma - \varpi^2)R$ *and* in $\varsigma\phi^2$; three independent occurrences, so no single renaming removes it. That is precisely why ς is a physical parameter and υ_2 is not.

Practical consequence. υ_2 and H_0 enter only as H_0^2/υ_2 and are therefore exactly degenerate. They must never be sampled together. In the pipeline υ_2 is fixed (`MUPS2_FIXED`), which is necessary rather than merely convenient. υ_2 becomes physical at perturbation level, where a comoving wavenumber k introduces a scale that does not co-rescale with τ.

5 Consequence 2: only ratios of the Ω's are measurable

The symmetry (5) multiplies all three Ω's by a common factor. Any function of $(\Omega_{r,0}, \Omega_{m,0}, \Omega_{\Lambda,0})$ that survives in $E(z)$ must therefore be invariant under that common rescaling, i.e. must be a function of the ratios alone. Explicitly, factoring (4),

$$S(a) = \Omega_{r,0}\left[a^{-4} + \frac{\Omega_{m,0}}{\Omega_{r,0}}a^{-3} + \frac{\Omega_{\Lambda,0}}{\Omega_{r,0}}\right], \tag{10}$$

the prefactor $\Omega_{r,0}$ sets the *height* of the source and is absorbed by (7); the bracket sets its *shape* and depends on two ratios only. The shape is fixed by the two crossover epochs

$$\Omega_{r,0}a^{-4} = \Omega_{m,0}a^{-3} \implies 1 + z_{\mathrm{eq}} = \frac{\Omega_{m,0}}{\Omega_{r,0}}, \qquad \Omega_{m,0}a^{-3} = \Omega_{\Lambda,0} \implies 1 + z_{\Lambda} = \left(\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}\right)^{1/3}, \tag{11}$$

both manifestly ratios. Three input numbers thus carry two physical (shape) numbers plus one unobservable (height).

6 Consequence 3: the $(\Omega_{m,0}, \Omega_{\Lambda,0})$ wedge

In the fit $\Omega_{r,0}$ is held fixed, so the exact symmetry cannot be realised; only the two-parameter motion

$$(\Omega_{m,0}, \Omega_{\Lambda,0}) \to (\kappa^2\Omega_{m,0},\ \kappa^2\Omega_{\Lambda,0}) \tag{12}$$

is available. This differs from the exact symmetry only by the piece that was not rescaled,

$$\Delta S = (\kappa^2 - 1)\,\Omega_{r,0}a^{-4}, \tag{13}$$

which is the *sole* source of constraint along that direction. Over the redshift range covered by DESI DR2 BAO, $\Omega_{r,0}a^{-4}$ is below 0.1% of S (Table 2), so the symmetry is only weakly broken and the posterior stretches into a wedge pointing at the origin, with slope $\Omega_{\Lambda,0}/\Omega_{m,0}$ and angular width set by the uncertainty on that ratio.

7 Comparison with ΛCDM

The same symmetry exists in ΛCDM. From $H^2 = H_0^2(\Omega_{m,0}a^{-3} + \Omega_{r,0}a^{-4} + \Omega_{\Lambda,0})$, scaling all three Ω's by κ^2 gives $H \to cH$ and leaves $E = H/H_0$ unchanged. ΛCDM escapes the degeneracy only because flatness,

$$\Omega_{m,0} + \Omega_{r,0} + \Omega_{\Lambda,0} = 1, \tag{14}$$

is a relation among the Ω's alone and forbids $\kappa \neq 1$. Equation (14) is simultaneously the Friedmann constraint at $a = 1$ and the normalisation $E(0) = 1$; it supplies a canonical denominator, so that $\Omega_{m,0} = \Omega_{m,0}/(\Omega_{m,0}+\Omega_{r,0}+\Omega_{\Lambda,0})$ identically and $\Omega_{m,0}$ may be read as a fraction.

TorC possesses a constraint at $a = 1$ as well, but it involves ϖ_0 and $\dot{\varpi}_0$ and does not reduce to a relation among the Ω's. No canonical denominator exists, so the normalisation must be fixed by convention — in the pipeline, by hardcoding $\Omega_{r,0} = 8.24 \times 10^{-5}$. Consequently $\Omega_{m,0}$ alone is a ratio with respect to an arbitrary reference, whereas

$$\frac{\Omega_{m,0}}{\Omega_{m,0} + \Omega_{r,0} + \Omega_{\Lambda,0}} \tag{15}$$

is invariant under (5) and is the correct analogue of the ΛCDM $\Omega_{m,0}$.

Both theories measure only ratios. The difference is that ΛCDM has a natural denominator and TorC does not.

8 Numerical verification

All checks use the pipeline solver (Dopri8, `rtol`=`atol`=10^{-8} unless stated); quoted residuals are at solver tolerance.

Table 1: The symmetry acting at the posterior-mean point, $\kappa^2 = 4$. The source is multiplied by κ^2 and H by c at *every* epoch, so E is unchanged.

a	S (κ=1)	S (κ^2=4)	ratio	H (κ=1)	H (κ^2=4)	ratio
10^{-6}	8.2693×10^{19}	3.3077×10^{20}	4.00	8.7004×10^{11}	1.7401×10^{12}	2.00000
10^{-4}	1.1167×10^{12}	4.4668×10^{12}	4.00	1.0342×10^{8}	2.0685×10^{8}	2.00000
10^{-2}	3.0094×10^{5}	1.2038×10^{6}	4.00	6.8963×10^{4}	1.3793×10^{5}	2.00000
10^{-1}	2.9453×10^{2}	1.1781×10^{3}	4.00	2.2076×10^{3}	4.4152×10^{3}	2.00000
0.5	3.3529	13.412	4.00	2.2513×10^{2}	4.5026×10^{2}	2.00000
1.0	1.3028	5.2111	4.00	1.2582×10^{2}	2.5163×10^{2}	2.00000

The resulting $E(z)$ agree to 1.0000000 in ratio at every tabulated a; across $\kappa^2 \in [0.5, 10]$ the maximum deviation is $\max|E/E_0 - 1| \sim 10^{-9}$. Holding the two ratios fixed while varying the overall scale by a factor 10^4 ($\Omega_{m,0}$ from 0.003 to 30) gives $\max|E/E_0 - 1| = 4.4 \times 10^{-9}$.

Table 2: Radiation's share of the source over the DESI DR2 BAO range. This is the only term breaking the $(\Omega_{m,0}, \Omega_{\Lambda,0})$ degeneracy.

z	$\Omega_{r,0}a^{-4}$	share of S
0.0	8.240×10^{-5}	0.0063%
0.5	4.172×10^{-4}	0.0209%
1.0	1.318×10^{-3}	0.0393%
2.0	6.674×10^{-3}	0.0748%
2.6	1.384×10^{-2}	0.0943%

Table 3: Cost in log-likelihood of moving along the degenerate direction. Scaling all three Ω's is exactly free; scaling only $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$ is almost free. For reference 1σ corresponds to $\Delta \ln \mathcal{L} \approx 0.5$.

κ^2	$\Delta \ln \mathcal{L}$, $\Omega_{m,0}, \Omega_{\Lambda,0}$ only	$\Delta \ln \mathcal{L}$, all three
0.70	-0.0063	2.6×10^{-8}
1.00	0.0000	0
1.50	-0.0179	2.4×10^{-8}
2.00	-0.0437	8.8×10^{-9}

υ_2**.** Rate-homogeneity applied to the second-order form of the equations implies the identity

$$f\big(a, \lambda\dot{a}, \varpi, \lambda\dot{\varpi},\ \upsilon_2/\lambda^2\big) = \lambda^2\, f\big(a, \dot{a}, \varpi, \dot{\varpi},\ \upsilon_2\big), \tag{16}$$

valid for any λ. Equation (16) holds to 7×10^{-12} and 9×10^{-15} for the two vector-field components over 2000 random points (symbolically exact, verified with SYMPY). The series initial conditions rescale consistently: with $\sigma = \tau/\sqrt{\upsilon_2}$, the values of $\sigma_{\rm ini}$, a, ϖ and $\sqrt{\upsilon_2}\,\mathrm{d}a/\mathrm{d}\tau$ agree across $\upsilon_2 \in [0.5, 3]$ to 1.6×10^{-16}. Measured H ratios reproduce $\upsilon_2^{-1/2}$ to 6×10^{-10}, and $\max|E/E_0 - 1| \leq 1.9 \times 10^{-9}$.

ΛCDM. As implemented with flatness, scaling $\Omega_{m,0}$ and $\Omega_{r,0}$ together costs $\Delta \ln \mathcal{L} = -2472$ and -1838 for $\kappa^2 = 0.5, 2$. Freeing $\Omega_{\Lambda,0}$ and normalising by $E(0)$ restores the symmetry exactly: $\Delta \ln \mathcal{L} = 0$ for κ^2 up to 10.

9 Summary of measured quantities

The invariant $\Omega_{m,0}/(\Omega_{m,0}+\Omega_{r,0}+\Omega_{\Lambda,0}) = 0.2216 \pm 0.0477$ is both better determined than $\Omega_{m,0}$ itself (21% versus 45%, since taking the ratio removes the spread along the wedge) and directly comparable with the ΛCDM value 0.2972 ± 0.0087, from which it differs by $\approx 1.6\sigma$.

Table 4: DESI DR2 BAO, 1000 live points. Invariant combinations are measurable; $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$ individually inherit the convention $\Omega_{r,0} = 8.24 \times 10^{-5}$ and are additionally limited by the prior along the wedge (7.3% of the posterior mass lies within 0.05 of the $\Omega_{\Lambda,0}$ prior ceiling of 1.5).

quantity	value	status
$\Omega_{m,0}/\Omega_{r,0} = 1 + z_{\rm eq}$	3552 ± 1606	invariant
$\Omega_{\Lambda,0}/\Omega_{m,0}$	3.75 ± 1.13	invariant (wedge slope)
$z_\Lambda = (\Omega_{\Lambda,0}/\Omega_{m,0})^{1/3} - 1$	0.539 ± 0.150	invariant
$\Omega_{m,0}/(\Omega_{m,0}+\Omega_{r,0}+\Omega_{\Lambda,0})$	0.2216 ± 0.0477	invariant
$\Omega_{m,0}$	0.2927 ± 0.1323	convention-dependent
$\Omega_{\Lambda,0}$	1.0100 ± 0.3475	convention-dependent
ΛCDM $\Omega_{m,0}$	0.2972 ± 0.0087	invariant (flatness)